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AD-GBC: Anisotropic Granular-Ball Skip-Connection Refiner for UNet-Based Medical Image Segmentation

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Abstract

Prototype or region-attention modules have recently improved medical image segmentation but still suffer from two fundamental limitations: 1) they represent each semantic concept as a point or isotropic region, failing to capture the inherently anisotropic geometry of real feature distributions; and 2) many rely on non-differentiable clustering or one-way kernel weighting, which restricts their ability to form coherent region-level representations. We address these issues with the Anisotropic Differentiable Granular-Ball (AD-GBC) module, which generalizes prototypes into learnable geometric regions parameterized by a center and an anisotropic vector scale. AD-GBC aggregates local features into region-level semantics and redistributes the refined representation back to pixels in a fully differentiable manner, enabling geometry-aware refinement within modern UNet-style architectures. Two geometric regularizers, a Wasserstein-based diversity loss and a scale consistency loss, mitigate center collapse and encourage stable, well-formed region geometry. AD-GBC yields consistent improvements across four widely used medical segmentation benchmarks (BUSI, GlaS, CVC-ClinicDB, ISIC17) when integrated into two strong backbones (Rolling-UNet and U-KAN), demonstrating that the proposed geometric region formulation generalizes well across different imaging conditions. The code is available at https://github.com/SiaShen-dot/AD-GBC.

1. Introduction

Semantic segmentation of medical images is crucial for clinical diagnostics, enabling quantitative analysis of structures such as tumors and lesions [3, 21]. The UNet architecture, with its encoder–decoder structure and skip connections, has become the de-facto standard for this task [8, 21]. However, the naive concatenation used in standard skip connections often introduces semantic inconsistency and noise [19, 24], especially in challenging cases with fuzzy lesion boundaries and cluttered background textures. To address this, recent works move beyond plain feature concatenation and introduce prototype- or slot-based refinement mechanisms on top of encoder–decoder backbones. Methods such as ProtoSeg [26] and Slot Attention [10, 15] reinterpret skip fusion as interactions between pixel features and a compact set of learnable prototypes (or "slots"). These prototypes act as semantic anchors that help organize raw features into more consistent modes, leading to better region consistency and boundary delineation [19, 26].

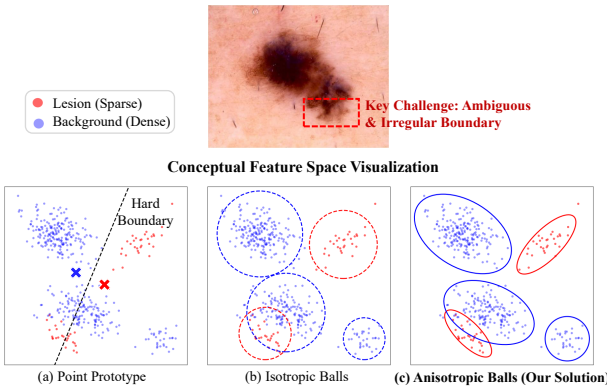


Figure 1. Motivation. (a) Point prototypes fail on multi-mode, irregular regions. (b) Isotropic balls fit poorly with scalar radii. (c) Our module models each semantic mode with anisotropic region geometry.

Yet, despite acting as powerful skip refiners, these methods still model each semantic concept as a *point* in feature space and rely on one-way or iterative attention without any explicit control over region geometry. Consequently, they struggle to capture the *multi–mode feature distributions, anisotropic shapes, and overlapping patterns* that characterize real lesion regions with ambiguous appearance and irregular boundaries, as illustrated in Fig. 1.

1) Multi-Mode Feature Distributions. A single semantic class (e.g., "Lesion") is often not a single cluster but is multi-mode in the feature space, comprising multiple dis-

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tinct visual modes (e.g., a dark core vs. inflamed edges). We provide visual proof of this multi-mode structure in the Supplementary Material. As shown in Fig. 1(a), traditional point-prototype methods [19, 26], especially those using $K = M$ (K prototypes for M classes), fundamentally fail to model this multi-mode distribution [19].

2) Anisotropic Feature Geometry. While more advanced methods like Granular-Ball Computing (GBC) [28] use $K \gg M$ balls to solve the multi-mode problem, they rely on a critical, flawed assumption: that the feature distribution of each mode is isotropic (i.e., representable by a scalar radius, or a "circle"). As visualized in Fig. 1(b), we argue that real-world feature clusters are inherently anisotropic (i.e., "elliptical").

3) Ambiguity (Overlapping Features) and Class Imbalance. This fundamental geometric mismatch, the conflict between anisotropic data (Fig. 1(b), the data) and isotropic models (Fig. 1(b), the dashed circles), is catastrophically amplified by two other challenges: semantic ambiguity (i.e., overlapping features) and class imbalance (dense background vs. sparse lesion). When an isotropic (scalar-ball) prototype attempts to model anisotropic data, it faces a disastrous trade-off: it must become "oversized" to cover the cluster's long axis (as shown in Fig. 1(b)). This oversized prototype, representing the sparse "lesion" class, inevitably creates a massive, false overlap with the surrounding dense "background" clusters. This flawed geometry, a direct result of the isotropic assumption, leads to a catastrophic failure in modeling the true, subtle ambiguity at the class boundaries.

To address these issues, we introduce the Anisotropic Differentiable Granular-Ball (AD-GBC) module. AD-GBC replaces point prototypes with learnable anisotropic regions parameterized by a vector scale $\boldsymbol{\sigma}_k \in \mathbb{R}^D$ (Fig. 1(c)), allowing each anchor to model the true shape and extent of the feature clusters. The module aggregates pixel features into region-level representations and feeds the refined semantics back to the pixels in a fully differentiable manner. We further integrate AD-GBC into a UNet-based model using a weight-shared semantic skip connection (Fig. 2, top), ensuring consistent and geometry-aware feature alignment before fusion. Our contributions are three-fold:

- We propose the AD-GBC module, which generalizes point prototypes into anisotropic, bounded geometric regions and introduces two coupled geometric regularizers ($\mathcal{L}_{\text{W-div}}$, $\mathcal{L}_{\text{scale-con}}$) to prevent anchor collapse and enforce consistent region geometry.
- We design a weight-shared semantic skip-connection refiner that integrates AD-GBC into UNet-based architectures, refining semantic alignment before feature fusion.
- Across four medical segmentation benchmarks (BUSI, GlaS, CVC-ClinicDB, ISIC17), AD-GBC brings consistent improvements to strong backbones (RollingUNet [14], U-KAN [13]), demonstrating the effectiveness and robustness of the proposed geometric region formulation.

Table 1. Conceptual comparison highlighting the transition from point anchors to anisotropic region representations.

Method	Anchor Type	Spatial Support	Geometry
Gaussian Attn. [16]	point	implicit kernel	unbounded
Slot Attn. [10, 15]	point slots	implicit	none
Original GBC [9, 27–29]	center	explicit region	isotropic
AD-GBC (ours)	center	explicit region	anisotropic

2. Related Work

Encoder–Decoder Architectures. U-Net [20] and its variants form the foundation of medical image segmentation. Methods such as UNet++ [31, 32] enhance multi-scale fusion through nested skip connections, while pyramid-based designs [2, 12, 25] further improve semantic–spatial integration. These architectures provide strong feature extraction and multi-scale integration, but they typically rely on pixel-wise feature representations without explicit region modeling.

Prototype- and Slot-Based Segmentation. Prototype-based methods [30] represent semantic structures using point anchors with soft assignments. Gaussian Attention [16] introduces covariance-weighted kernels to model spatial influence around anchor points, while Slot Attention [10, 15] iteratively updates slot representations to capture object-centric structure. However, these approaches primarily rely on point-based anchors with implicit spatial support rather than explicitly modeling bounded regions.

Granular-Ball Computing (GBC). GBC [9, 27–29] represents data using local balls defined by a center and radius, enabling multi-granularity modeling of the data space. However, existing GBC methods rely on non-differentiable clustering procedures and assume isotropic regions, which limits their integration into end-to-end deep neural networks.

Table 1 summarizes the conceptual differences between existing prototype-based methods and our approach. While Gaussian attention and slot-based methods rely on point anchors with implicit spatial support, and classical GBC models isotropic regions, our method introduces an anisotropic region-ball representation that is fully differentiable and compatible with end-to-end training.

3. Method

As illustrated in Fig. 2, our method proposes the Anisotropic Differentiable Granular-Ball Computing (AD-GBC) module, which is integrated as a novel semantic refiner within a UNet architecture to solve the geometric flaws

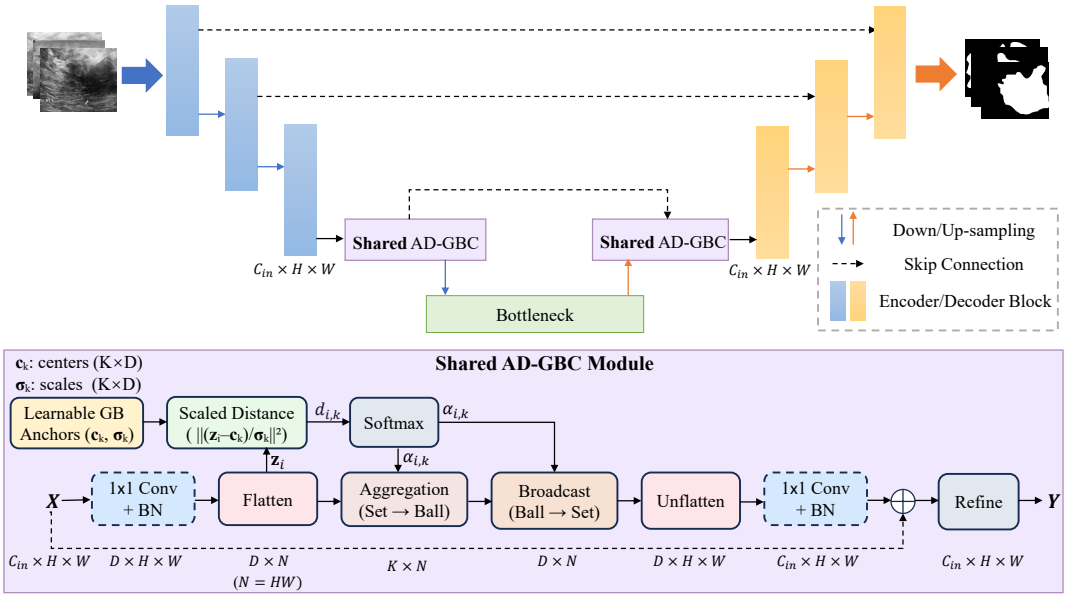


Figure 2. Conceptual overview of the AD-GBC module integration. Refinement is illustrated at skip level 3. The two AD-GBC modules *share weights* to enforce consistent semantic refinement across encoder and decoder pathways.

identified in Sec. 1.

This section follows a top-down structure. We first describe the overall network architecture and our proposed integration strategy (Sec. 3.1). We then detail the core AD-GBC module itself (Sec. 3.2), followed by the geometric regularization losses that enable its functionality (Sec. 3.3), the final composite objective (Sec. 3.4), and a computational complexity analysis (Sec. 3.5).

3.1. Overall Architecture: AD-GBC as a Semantic Skip-Connection Refiner

The UNet architecture [20] is the standard for medical image segmentation due to its powerful skip-connection mechanism. These connections fuse deep, semantic features from the encoder with shallow, high-resolution features from the decoder, enabling precise localization. However, standard skip connections typically perform a naive concatenation or addition. This can be suboptimal, as the deep encoder features, while semantic, may also contain noise or irrelevant context that contaminates the high-fidelity decoder pathway.

To address this, we propose integrating our AD-GBC module not as a simple bottleneck replacement, but as a Semantic Skip-Connection Refiner. The goal is to ensure that the features fused across the skip connection are not only high-level, but are also semantically aligned and purified by our AD-GBC module.

Integration Strategy As conceptually illustrated in Fig. 2, we deploy the AD-GBC module onto the deeper, more semantic skip-connection pathways of the UNet (e.g., at the third encoder/decoder level). This integration follows a dual-path, shared-weight strategy:

- Encoder Path Refinement: The feature map from the encoder (e.g., from Encoder Block 3) is first passed through the "Shared AD-GBC" module. This step refines the encoder's raw semantic features, projecting them onto the K learnable granular balls.
- Decoder Path Refinement: Concurrently, the corresponding feature map being upsampled by the decoder (e.g., from Decoder Block 4) is passed through the *exact same* "Shared AD-GBC" module.
- Semantic Fusion: The refined encoder and decoder features are fused (e.g., via element-wise addition), forming a shared semantic representation used as the skip connection for the next decoder block (e.g., Decoder Block 3).

This weight-shared strategy forces both encoder and decoder features to be projected onto the same learned semantic manifold before fusion, ensuring a high degree of semantic consistency. The detailed mechanism of how the AD-GBC module achieves this purification is described next.

3.2. Anisotropic Differentiable Granular-Ball Computing (AD-GBC) Module

Traditional prototype-based modules represent semantics using a set of learnable points. While such "point prototypes" work well for global classification, they oversimplify segmentation targets whose regions exhibit smooth appearance transitions and ambiguous boundaries. This motivates a transition from *point-based anchoring* to *soft region anchoring*, where each concept is modeled as a feature-space

region rather than a single point.

GBC embodies this regional perspective through granular balls and has demonstrated strong potential for adaptive multi-scale representation. Yet, its classical formulation remains non-differentiable and depends on manually defined radii, preventing seamless integration into gradient-based deep networks.

To address this, we propose the *Anisotropic Differentiable Granular-Ball Computing (AD-GBC)* module. Each granular ball is treated as a learnable *soft region anchor* parameterized by a center and an anisotropic radius (scale), and interacts with pixel features through a differentiable *Set→Ball→Set* pathway: features are softly assigned to balls, aggregated into region-level consensus, and then broadcast back to refine the original feature map. All parameters of the balls (centers and radii) are learned in a fully data-driven, end-to-end manner, allowing AD-GBC to serve as a plug-and-play regional geometry layer in UNet-like architectures.

3.2.1. Module Overview

As illustrated in Fig. 2, the AD-GBC module is designed to refine an input feature map $\mathbf{X} \in \mathbb{R}^{B\times C_{in}\times H\times W}$ into an output map $\mathbf{Y}$ of the same dimensions, which possesses richer contextual information.

The input $\mathbf{X}$ first undergoes feature projection and flattening. The core of the module is a two-stage interaction, Aggregation (Set→Ball) and Broadcast (Ball→Set), which operates in the flattened feature space. This interaction is mediated by a set of K learnable Granular-Ball (GB) Anchors. These anchors, which are central to the module's function, are parameterized by:

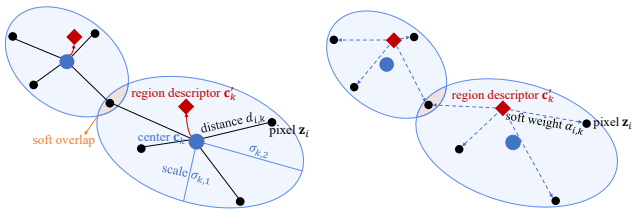
- *Anchor locations (centers):* A set $\mathbf{C} = \{\mathbf{c}_1, \ldots, \mathbf{c}_K\} \in \mathbb{R}^{K\times D}$, representing semantic locations.
- *Anisotropic Anchor Scales (Radii):* $\mathbf{\Sigma} = \{\boldsymbol{\sigma}_1, \ldots, \boldsymbol{\sigma}_K\}$ defines the soft extent of each anchor in the D-dimensional feature space. Conventional isotropic scales use a scalar radius ($\boldsymbol{\sigma}_k \in \mathbb{R}$), enforcing a hyperspherical shape. We instead adopt an *anisotropic* vector scale $\boldsymbol{\sigma}_k \in \mathbb{R}^D$ (constrained positive via Softplus), allowing each anchor to form a learnable hyper-ellipsoid in feature space.

Finally, the processed features are unflattened, fused with the original input via a residual connection, and passed through a refinement block to produce $\mathbf{Y}$.

3.2.2. Forward Computation

The forward computation, following the path in Fig. 2, proceeds in three main stages:

1) Feature Projection and Flattening: Given that the AD-GBC anchors operate in a D-dimensional space, the input feature map $\mathbf{X}$ is first optionally projected. A lightweight 1×1 Convolution (Conv) followed by Batch Normalization (BN), f_{proj}, aligns the channel dimension



(a) Aggregation (Set→Ball) (b) Broadcast (Ball→Set)

Figure 3. Visualization of the AD-GBC process.

C_{in} to D if they are different.

$$\mathbf{Z} = \begin{cases} f_{\text{proj}}(\mathbf{X}), & C_{in} \neq D, \\ \mathbf{X}, & C_{in} = D. \end{cases} \tag{1}$$

The resulting map $\mathbf{Z} \in \mathbb{R}^{B\times D\times H\times W}$ is then flattened into a sequence of $N = HW$ pixel features $\mathbf{Z}_{\text{flat}} \in \mathbb{R}^{B\times N\times D}$, which we denote as $\{\mathbf{z}_i\}_{i=1}^N$.

2) Core AD-GBC Interaction (Set ↔ Ball): This stage involves the bidirectional information flow between the pixel features (the "Set") and the GB anchors (the "Ball"). This process is conceptually visualized in Fig. 3.

First, we compute the soft membership of each pixel feature $\mathbf{z}_i$ to each GB anchor $(\mathbf{c}_k, \boldsymbol{\sigma}_k)$. This is not based on simple similarity but on a *scaled distance*. This distance measures how many "radii" ($\boldsymbol{\sigma}_k$) a pixel is from a center ($\mathbf{c}_k$):

$$d_{i,k} = \|(\mathbf{z}_i - \mathbf{c}_k) \oslash \boldsymbol{\sigma}_k\|_2^2, \tag{2}$$

where $\oslash$ denotes element-wise division. These distances are then normalized via a 'Softmax' (with temperature τ) to produce fuzzy assignment weights $\alpha_{i,k}$, representing the soft weight of pixel $\mathbf{z}_i$ belonging to ball k:

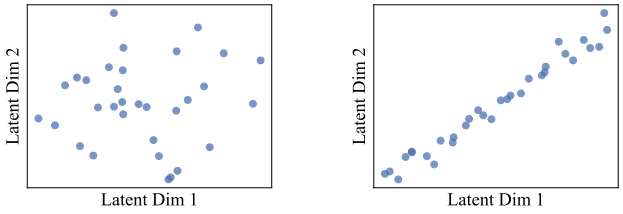
$$\alpha_{i,k} = \frac{\exp(-d_{i,k}/\tau)}{\sum_{j=1}^K \exp(-d_{i,j}/\tau)}. \tag{3}$$

With these weights, $\alpha_{i,k}$, the interaction proceeds in two steps:

- *Aggregation (Set→Ball):* As visualized in Fig. 3(a), the module gathers information from the pixel "Set" to form new region descriptors $\mathbf{c}'_k$. Each descriptor is a weighted sum of all pixel features, summarizing the consensus of the features belonging to that soft region:

$$\mathbf{c}'_k = \sum_{i=1}^N \alpha_{i,k}\mathbf{z}_i. \tag{4}$$

- *Broadcast (Ball→Set):* As shown in Fig. 3(b), the module then broadcasts this learned region-level context back to the pixel "Set". Each pixel's refined representation, $\hat{\mathbf{z}}_i$, is constructed as a weighted sum of the region descriptors,



(a) High-Rank (Diverse) (b) Low-Rank Collapse

Figure 4. Conceptual illustration of GB center distributions. (a) A high-rank, diverse set of centers spans the semantic space. (b) A low-rank, collapsed set of centers fails to capture diverse modes.

using the same weights $\alpha_{i,k}$:

$$\hat{\mathbf{z}}_i = \sum_{k=1}^{K} \alpha_{i,k} \mathbf{c}'_k. \tag{5}$$

3) Residual Fusion and Refinement: The sequence of refined pixel features $\{\hat{\mathbf{z}}_i\}$ is 'Unflattened' back into a feature map $\hat{\mathbf{Z}} \in \mathbb{R}^{B\times D\times H\times W}$. This map is passed through an optional 1×1 Conv followed by BN, denoted as $\tilde{f}_{\text{proj}}$, to project its dimension D back to C_{in}, yielding $\hat{\mathbf{X}}$.

Following the residual path in Fig. 2, this context-aware map $\hat{\mathbf{X}}$ is added back to the original input $\mathbf{X}$. This connection stabilizes training and preserves the original local details. The fused map is finally passed through a lightweight 'Refine' block, f_{refine} (a 3×3 Conv-BN-ReLU (Rectified Linear Unit) sequence), to enhance local consistency. The final module output $\mathbf{Y}$ is:

$$\mathbf{Y} = f_{\text{refine}}(\mathbf{X} + \hat{\mathbf{X}}). \tag{6}$$

Note that when the anisotropic radius $\boldsymbol{\sigma}_k$ degenerates to a scalar and the consensus step is removed, AD-GBC reduces to the standard Gaussian kernel weighting [16].

3.3. Geometric Regularization for AD-GBC

The AD-GBC module introduces learnable anchors $\{\mathbf{c}_k\}$ and anisotropic scales $\{\boldsymbol{\sigma}_k\}$ to represent semantic regions. However, when trained only with the task loss, these parameters may become degenerate, leading to collapsed anchors or unstable scale estimates. To ensure meaningful geometric representations, we introduce two complementary regularization losses.

3.3.1. Diversity Regularization for GB Centers

In the AD-GBC module, the K GB centers $\{\mathbf{c}_k\}$ act as learnable semantic anchors. To ensure that the *Set→Ball→Set* process is expressive, these anchors must be diverse and specialize in distinct visual modes. If multiple centers collapse into a low-dimensional subspace, they fail to represent distinct semantic modes. We refer to this failure as *semantic collapse*. As illustrated in Fig. 4(b), this causes the 'Broadcast' step to degenerate into trivial mixing, severely damaging both the module's representational power and its interpretability.

A Heuristic Approach and Its Limitations: A common heuristic for encouraging diversity is to enforce pairwise dissimilarity. We can adopt a penalty based on the squared cosine similarity, which encourages mutual orthogonality:

$$\mathcal{L}_{\text{cos-div}}(\mathbf{C}) = \frac{1}{K(K-1)} \sum_{j\neq k} \left(\frac{\mathbf{c}_j^\top \mathbf{c}_k}{\|\mathbf{c}_j\|_2 \, \|\mathbf{c}_k\|_2} \right)^2, \tag{7}$$

where $\mathbf{C} \in \mathbb{R}^{K\times D}$ is the set of K GB centers.

However, the loss $\mathcal{L}_{\text{cos-div}}$ in Eq. (7) cannot regulate the dimensionality of the subspace they span. This limitation is formalized in Theorem 1.

Theorem 1 (Cosine Loss Cannot Detect Semantic Collapse)**.** *Let $\mathbf{C} \in \mathbb{R}^{K\times D}$ be nonzero centers and $G(\mathbf{C})$ their normalized Gram matrix. If $\operatorname{rank}(G(\mathbf{C})) = r < K$, then there exists $\widetilde{\mathbf{C}} \in \mathbb{R}^{K\times r}$ such that $G(\widetilde{\mathbf{C}}) = G(\mathbf{C})$, which implies $\mathcal{L}_{\text{cos-div}}(\widetilde{\mathbf{C}}) = \mathcal{L}_{\text{cos-div}}(\mathbf{C})$.*

Proof. Please refer to the Supplementary Material. □

A Principled Solution (Wasserstein-based Uniformity): To address the limitations of pairwise metrics, we regularize the entire distribution of the K centers rather than only their pairwise relations. Inspired by the Wasserstein uniformity formulation in [11], we adopt the following loss:

$$\mathcal{L}_{\text{W-div}}(\mathbf{C}) = \underbrace{\|\hat{\mu}_C\|_2^2}_{\text{Centering}} + \underbrace{\operatorname{tr}(\hat{\Sigma}_C) - \frac{2}{\sqrt{D}} \operatorname{tr}\left(\hat{\Sigma}_C^{1/2}\right)}_{\text{Spectral uniformity regularization}}, \tag{8}$$

where $\hat{\mu}_C$ and $\hat{\Sigma}_C$ are the empirical mean and covariance matrix of the centers: $\hat{\mu}_C = \frac{1}{K}\sum_{k=1}^{K} \mathbf{c}_k$, $\hat{\Sigma}_C = \frac{1}{K}\sum_{k=1}^{K}(\mathbf{c}_k - \hat{\mu}_C)(\mathbf{c}_k - \hat{\mu}_C)^\top$.

This formulation encourages a well-spread and high-rank representation in feature space. We provide a theoretical analysis of this constrained optimum in the Supplementary Material.

In practice, this regularizer promotes a diverse set of anchors, as illustrated in Fig. 4(a).

3.3.2. Scale Consistency Regularization for Anisotropy

The anisotropic (vector) scale $\boldsymbol{\sigma}_k \in \mathbb{R}^D$ is central to AD-GBC, but task loss alone provides only a weak supervisory signal, making $\boldsymbol{\sigma}_k$ prone to collapse or geometrically inconsistent values. To provide an explicit constraint, we introduce a Scale Consistency Loss $\mathcal{L}_{\text{scale-con}}$.

Observed Anisotropic Scale $\mathbf{s}_k$ (The Target): We first define the observed anisotropic variance, $\mathbf{s}_k^2$, which is a $\mathbb{R}^D$

vector representing the geometric spread (variance) of the features assigned to the ball along each dimension:

$$\mathbf{s}_k^2 = \frac{\sum_{i=1}^{N} \alpha_{i,k} \cdot (\mathbf{z}_i - \mathbf{c}_k)^{\circ 2}}{(\sum_{i=1}^{N} \alpha_{i,k}) + \epsilon}, \tag{9}$$

where $\alpha_{i,k}$ is the soft membership (from Eq. 3), $(\cdot)^{\circ 2}$ is element-wise squaring, and ϵ is a small constant (e.g., 10^{-6}) for numerical stability to prevent division by zero. The *observed anisotropic scale* (standard deviation) is then the element-wise square root of this variance $\mathbf{s}_k = \sqrt{\mathbf{s}_k^2}$.

The Scale Consistency Loss $\mathcal{L}_{\text{scale-con}}$ (The Constraint): Finally, the $\mathcal{L}_{\text{scale-con}}$ minimizes the Euclidean distance (L2) between the observed variance ($\mathbf{s}_k^2$, treated as the target) and the learned variance ($\boldsymbol{\sigma}_k^2$, treated as the prediction):

$$\mathcal{L}_{\text{scale-con}} = \frac{1}{K} \sum_{k=1}^{K} \left\| \text{detach}(\mathbf{s}_k^2) - \boldsymbol{\sigma}_k^2 \right\|_2^2. \tag{10}$$

We compare the variances ($\mathbf{s}_k^2$ vs. $\boldsymbol{\sigma}_k^2$) rather than the scales ($\mathbf{s}_k$ vs. $\boldsymbol{\sigma}_k$) to improve numerical stability and avoid the $\sqrt{\cdot}$ operation, whose gradient may explode when the observed variance approaches zero.

The detach$(\cdot)$ operator prevents a circular dependency, $\boldsymbol{\sigma}_k$ affects $\alpha_{i,k}$, which affects $\mathbf{s}_k^2$, ensuring a stable one-way constraint that drives the learned scale toward the observed one.

3.4. Composite Loss Function

Our model is trained end-to-end with a composite loss function that combines the primary segmentation loss $\mathcal{L}_{\text{task}}$ (e.g., a combination of Cross-Entropy and Dice loss) with our two geometric regularizers. These are the Wasserstein-based diversity loss $\mathcal{L}_{\text{W-div}}$ (Eq. (8)) and the Scale Consistency Loss $\mathcal{L}_{\text{scale-con}}$ (Eq. (10)). The final objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{W-div}} + \beta \mathcal{L}_{\text{scale-con}}, \tag{11}$$

where λ and β are hyperparameters that balance the primary task objective with the diversity and scale consistency regularization, respectively.

3.5. Complexity and Efficiency Comparison

We analyze the efficiency of the AD-GBC module in terms of computational cost (measured in floating point operations, FLOPs) and parameter count. Let $N = HW$ be the number of pixels, K the number of granular balls, D the projection dimension, and C_{in} the input channels. The detailed, step-by-step breakdown of this analysis is provided in the Supplementary Material.

The module's full computational cost is determined by the projection layers, the core GBC interaction, and the final refinement block. All steps scale linearly with N, resulting in a *total complexity* of $\mathcal{O}(N \cdot (C_{in}D + KD + C_{in}^2))$. The module's *parameter count*, $\mathcal{O}(KD + C_{in}D + C_{in}^2)$, is fixed and independent of the input resolution N. This makes the AD-GBC module both parameter-efficient and highly scalable for high-resolution images.

4. Experiments

Datasets. We evaluate our method on four public benchmarks spanning ultrasound, histopathology, colonoscopy, and dermoscopy imaging. BUSI [1]: 647 breast ultrasound images with tumor annotations, resized to 256×256. GlaS [22]: 165 histopathology images with gland masks, resized to 512×512. CVC-ClinicDB [4]: 612 colonoscopy frames with polyp masks, resized to 256×256. ISIC17 [7]: The ISIC 2017 training set containing 2,000 dermoscopy images with lesion masks, resized to 256×256.

Evaluation Metrics. We report three widely used metrics for medical image segmentation: Dice/F1 and Intersection-over-Union (IoU) for region-level accuracy, and the 95% Hausdorff Distance (HD95) for boundary-sensitive evaluation.

Implementation Details. We follow protocols [13, 14] with the same settings, including an 80%/20% random train/validation split, repeated over three different random data seeds, 400 fixed epochs without early stopping, and model selection by best validation IoU. All models are implemented in PyTorch and trained on a single NVIDIA A100 GPU with 32 GB memory. To validate the generality of our AD-GBC module, we integrate it into two backbones: Rolling-UNet [14] and U-KAN [13]. Training is conducted for 400 epochs using Adam with an initial learning rate of 1×10^{-4} (batch size $= 8$), while GBC parameters use a separate learning rate of 1×10^{-2}. We set the number of granular balls to $K = 32$, and the feature dimension D matches C_{in} without additional projection. The task loss $\mathcal{L}$task is a combination of binary cross-entropy (BCE) and Dice loss, and the loss weights are $\lambda \in \{0.01, 0.1\}$ for $\mathcal{L}_{\text{W-div}}$ and $\beta \in \{0.1, 0.05\}$ for $\mathcal{L}_{\text{scale-con}}$.

4.1. Performance Comparison

Main Results. We conduct a comprehensive quantitative evaluation against state-of-the-art (SOTA) methods and multiple backbones. The results are presented in **Table 2** (performance per dataset) and **Table 3** (efficiency and average performance).

The results demonstrate two key findings. First, as shown in Table 2, our method achieves highly competitive performance across the four datasets. Our complete model (Rolling-UNet-L + AD-GBC) attains the best on GlaS, CVC-ClinicDB, and ISIC17, while the U-KAN + AD-GBC variant achieves the highest scores on BUSI. Second, our AD-GBC module proves to be a general and effective plug-and-play component. The performance gains

Table 2. Quantitative comparison on BUSI, GlaS, CVC-ClinicDB, and ISIC17. For each backbone, the row below reports the performance after integrating the proposed AD-GBC module. Results are averaged over three runs, with standard deviations shown. Best results are in bold.

Methods	BUSI [1]		GlaS [22]		CVC [4]		ISIC17 [7]	
	IoU↑	F1↑	IoU↑	F1↑	IoU↑	F1↑	IoU↑	F1↑
U-Net (2015) [20]	64.75$_{\pm 1.41}$	77.43$_{\pm 1.30}$	87.26$_{\pm 0.67}$	92.82$_{\pm 0.56}$	83.88$_{\pm 0.73}$	91.15$_{\pm 0.35}$	77.38$_{\pm 0.51}$	87.19$_{\pm 0.33}$
Att-UNet (2018) [18]	66.36$_{\pm 1.65}$	79.22$_{\pm 1.55}$	86.98$_{\pm 1.71}$	92.89$_{\pm 0.89}$	84.70$_{\pm 0.47}$	91.55$_{\pm 0.25}$	76.24$_{\pm 0.23}$	86.53$_{\pm 0.10}$
U-Net++ (2018) [31]	66.51$_{\pm 1.88}$	79.34$_{\pm 1.78}$	87.74$_{\pm 0.72}$	92.96$_{\pm 0.49}$	84.52$_{\pm 1.44}$	91.50$_{\pm 0.87}$	78.72$_{\pm 1.01}$	86.20$_{\pm 0.76}$
U-NeXt (2022) [23]	61.47$_{\pm 1.51}$	74.67$_{\pm 1.03}$	84.18$_{\pm 0.71}$	91.55$_{\pm 0.75}$	74.58$_{\pm 0.27}$	85.21$_{\pm 0.18}$	82.16$_{\pm 0.06}$	90.07$_{\pm 0.11}$
U-Mamba (2024) [17]	65.89$_{\pm 2.98}$	78.55$_{\pm 3.02}$	87.48$_{\pm 0.42}$	93.02$_{\pm 0.33}$	84.35$_{\pm 0.63}$	91.63$_{\pm 0.37}$	81.23$_{\pm 0.32}$	89.33$_{\pm 0.42}$
Rolling-UNet-S (2024) [14]	65.38$_{\pm 2.96}$	78.34$_{\pm 2.19}$	86.05$_{\pm 0.49}$	92.71$_{\pm 0.07}$	82.87$_{\pm 1.40}$	90.48$_{\pm 0.85}$	82.14$_{\pm 1.33}$	90.22$_{\pm 0.90}$
+ AD-GBC	65.62$_{\pm 0.79}$	79.08$_{\pm 0.51}$	86.55$_{\pm 0.08}$	92.78$_{\pm 0.04}$	84.90$_{\pm 0.86}$	91.76$_{\pm 0.50}$	84.90$_{\pm 0.10}$	91.67$_{\pm 0.04}$
Rolling-UNet-M (2024) [14]	66.41$_{\pm 1.19}$	79.02$_{\pm 0.83}$	85.86$_{\pm 1.56}$	92.55$_{\pm 0.73}$	84.93$_{\pm 0.65}$	91.71$_{\pm 0.44}$	83.51$_{\pm 0.50}$	90.67$_{\pm 0.33}$
+ AD-GBC	67.36$_{\pm 0.97}$	80.42$_{\pm 0.70}$	88.16$_{\pm 0.25}$	93.70$_{\pm 0.14}$	87.07$_{\pm 0.86}$	93.04$_{\pm 0.50}$	84.88$_{\pm 0.09}$	91.64$_{\pm 0.04}$
Rolling-UNet-L (2024) [14]	67.85$_{\pm 1.76}$	80.22$_{\pm 1.14}$	87.99$_{\pm 0.31}$	93.47$_{\pm 0.29}$	85.77$_{\pm 0.40}$	92.11$_{\pm 0.23}$	84.10$_{\pm 0.38}$	91.10$_{\pm 0.21}$
+ AD-GBC	69.30$_{\pm 0.83}$	81.69$_{\pm 0.54}$	**89.37**$_{\pm 0.12}$	**94.38**$_{\pm 0.06}$	**87.45**$_{\pm 0.68}$	**93.25**$_{\pm 0.40}$	**84.95**$_{\pm 0.08}$	**91.72**$_{\pm 0.08}$
U-KAN (2025) [13]	65.76$_{\pm 1.45}$	79.03$_{\pm 1.27}$	87.65$_{\pm 0.31}$	93.47$_{\pm 0.06}$	84.94$_{\pm 0.64}$	91.78$_{\pm 0.39}$	84.52$_{\pm 0.44}$	91.43$_{\pm 0.26}$
+ AD-GBC	**69.86**$_{\pm 0.75}$	**82.12**$_{\pm 0.48}$	88.64$_{\pm 0.33}$	93.97$_{\pm 0.19}$	85.03$_{\pm 0.07}$	91.85$_{\pm 0.01}$	84.59$_{\pm 0.06}$	91.50$_{\pm 0.05}$

Table 3. Comparison of efficiency and average segmentation performance. The best results are bold.

Methods	Efficiency		Average Segmentation		Gains (Δ)	
	Params (M) ↓	GFLOPs ↓	IoU↑	F1↑	IoU↑	F1↑
U-Net (2015) [20]	31.04	524.2	78.32$_{\pm 0.83}$	87.15$_{\pm 0.64}$		
Att-UNet (2018) [18]	34.88	533.1	78.57$_{\pm 1.02}$	87.55$_{\pm 0.70}$		
U-Net++ (2018) [31]	36.63	1109.0	79.37$_{\pm 1.26}$	87.50$_{\pm 0.98}$		
U-NeXt (2022) [23]	1.47	4.58	75.60$_{\pm 0.64}$	85.38$_{\pm 0.52}$		
U-Mamba (2024) [17]	86.30	2087.0	79.74$_{\pm 1.09}$	88.13$_{\pm 1.04}$		
Rolling-UNet-S (2024) [14]	1.78	16.82	79.11$_{\pm 1.55}$	87.94$_{\pm 1.00}$		
Rolling-UNet-S + AD-GBC	1.82	19.91	80.49$_{\pm 0.46}$	88.82$_{\pm 0.27}$	+1.38	+0.88
Rolling-UNet-M (2024) [14]	7.10	66.25	80.18$_{\pm 0.98}$	88.49$_{\pm 0.58}$		
Rolling-UNet-M + AD-GBC	7.25	77.27	81.87$_{\pm 0.54}$	89.70$_{\pm 0.35}$	**+1.69**	**+1.21**
Rolling-UNet-L (2024) [14]	28.32	262.96	81.43$_{\pm 0.71}$	89.23$_{\pm 0.47}$		
Rolling-UNet-L + AD-GBC	28.92	304.32	**82.77**$_{\pm 0.43}$	**90.26**$_{\pm 0.27}$	+1.34	+1.04
U-KAN (2025) [13]	6.35	14.02	80.72$_{\pm 0.71}$	88.93$_{\pm 0.50}$		
U-KAN + AD-GBC	6.51	16.77	82.03$_{\pm 0.30}$	89.86$_{\pm 0.18}$	+1.31	+0.93

are consistent and clear. For instance, integrating AD-GBC boosts the U-KAN backbone by a substantial +4.10% IoU on BUSI (65.76 → 69.86).

Table 3 further shows that this performance gain comes at a minimal cost. The AD-GBC module introduces only ≈2% additional parameters (≈0.15-0.6M) and a modest ≈15-20% increase in GFLOPs. This confirms that our method achieves a highly favorable accuracy-to-cost trade-off, validating the efficiency of our anisotropic GB modeling.

Fig. 5 provides qualitative evidence for these quantitative gains. The visualizations highlight our model's ability to handle the ambiguous and irregular boundaries identified as a key challenge in our motivation (Fig. 1). More qualitative examples are provided in the Supplementary Material.

Interpretability via Anchor Visualization. Fig. 6 visualizes the soft assignments of selected anchors as heatmaps. Different anchors specialize in distinct semantic regions (e.g., lesion core vs. healthy skin), confirming that AD-GBC decomposes the feature space into interpretable regional components.

4.2. Ablation Study

To validate the design choices of our AD-GBC module, we conduct a thorough set of ablation studies on representative datasets (GlaS and BUSI). We analyze the contribution of our core components and the sensitivity to the number of granular balls, K.

Impact of Core Components and Losses. We evalu-

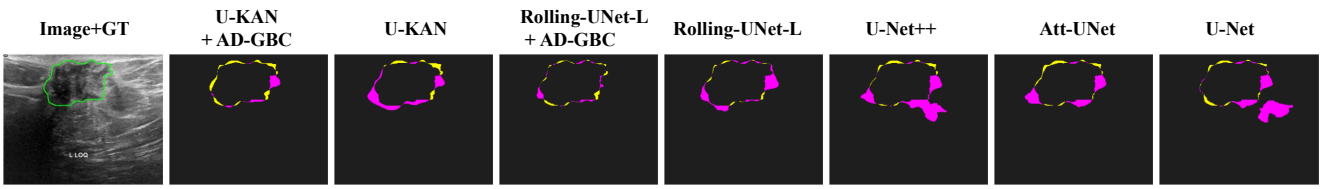
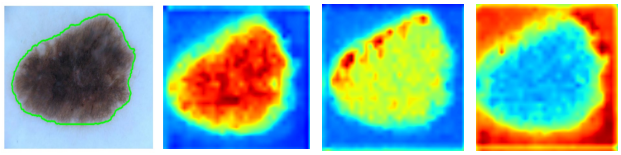


Figure 5. Qualitative comparison on a challenging sample from BUSI. The first panel is the original image, with the green contour indicating the Ground Truth. In the visualized segmentation results, purple indicates over-segmentation, and yellow indicates under-segmentation.



(a) Original + GT Boundary (b) Texture-Aware Anchor (c) Boundary-Sensitive Anchor (d) Background-Suppression Anchor

Figure 6. Representative GB anchors learned by the proposed AD-GBC module on ISIC17.

ate the necessity of three key components of AD-GBC—the anisotropic scale $\boldsymbol{\sigma}_k$, the diversity loss $\mathcal{L}_{\text{W-div}}$, and the scale consistency loss $\mathcal{L}_{\text{scale-con}}$—with results shown in Table 4.

Starting from the Rolling-UNet-L baseline (87.99% IoU), introducing a GBC module with an isotropic (scalar) scale brings only a marginal gain (+0.13%). Replacing it with our anisotropic (vector) scale increases performance to 88.28%, demonstrating the importance of modeling feature-space anisotropy.

On this anisotropic baseline, each geometric regularizer yields further improvements: $\mathcal{L}_{\text{W-div}}$ and $\mathcal{L}_{\text{scale-con}}$ raise accuracy to 88.70% and 88.78%, respectively, highlighting the roles of preventing anchor collapse and enforcing geometric consistency. Using both losses together produces a clear synergistic effect, achieving 89.63% IoU and forming our full model. Overall, all three components—anisotropic scales, diversity regularization, and scale consistency—are essential to AD-GBC's effectiveness.

Sensitivity to K, Temperature (τ), and Sparsity. Table 5 analyzes the sensitivity of AD-GBC to the number of centers K, temperature τ, and sparsity on BUSI. For moderate center sizes ($K = 16$ or 32), $\tau = 1.0$ performs best, while for larger $K = 64$, a smaller temperature ($\tau = 0.01$) becomes more effective, suggesting that lower τ helps sharpen assignments when the anchor set becomes dense. Sparsity also improves performance, particularly for large K; for example, when $K = 64$, restricting assignments to the top-5 anchors yields higher IoU than top-10. Overall, $K = 32$ with $\tau = 1.0$ provides a stable configuration.

Table 4. Ablation of the loss components on GlaS.

Configuration	$\mathcal{L}_{\text{W-div}}$	$\mathcal{L}_{\text{scale-con}}$	IoU↑	F1↑	HD95↓
Baseline (Rolling-UNet-L) [14]	×	×	87.99	93.47	0.65
+ GBC (Isotropic, scalar σ_k)	×	×	88.12	93.67	0.63
+ GBC (Anisotropic, vector $\boldsymbol{\sigma}_k$)	×	×	88.28	93.88	0.59
	√	×	88.70	94.00	0.56
	×	√	88.78	94.05	0.56
	√	√	**89.63**	**94.51**	**0.51**

Table 5. Sensitivity to temperature τ and sparsity (Top-5 / Top-10) on BUSI (IoU).

Setting	$K = 16$	$K = 32$	$K = 64$
$\tau = 0.01$	69.13	68.09	**69.65**
$\tau = 0.1$	66.76	69.90	69.16
$\tau = 1.0$	**69.58**	**70.71**	66.30
Top-5	69.26	70.46	**69.33**
Top-10	68.04	67.67	68.22

5. Discussion

Our results show that redefining prototypes as *anisotropic, bounded geometric regions* leads to clear improvements over isotropic formulations. As confirmed in Table 4, introducing anisotropic scales yields a noticeable increase in boundary accuracy (HD95 ↓ from 0.63 to 0.59). This geometry becomes effective only when properly stabilized: the two regularizers, $\mathcal{L}_{\text{W-div}}$ and $\mathcal{L}_{\text{scale-con}}$, act synergistically and jointly enable the full performance of AD-GBC (IoU from 88.78% to 89.63%).

Unlike Gaussian attention [16] and Slot Attention [10, 15], AD-GBC explicitly models bounded anisotropic regions and performs a consensus update between pixels and region anchors.

In addition, the current study focuses on CNN-based encoder–decoder architectures. Although the AD-GBC module is fully differentiable and could potentially be integrated into Transformer-based segmentation models (e.g., TransUNet [6] or Swin-UNet [5]) through pixel–prototype interactions, this direction remains to be explored in future work.

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